Literature Review and Dataset Collection for Robotic E-Waste Battery Extraction

1. Introduction

The problem of e-waste is a growing global issue due to its negative impact on the environment and potential hazards arising from the presence of lithium-ion batteries inside electronic items that may catch fire, release toxic chemicals and explode during the process of recycling. Removal of batteries manually using the traditional approach poses serious difficulties because of the inefficiency and hazards involved in the operation.

Recently developed techniques of computer vision, robotics, and artificial intelligence facilitate the creation of robots able to identify e-waste and safely remove batteries from electronic items automatically. Convolutional neural networks designed specifically for the object detection task have proven highly effective at recognizing different objects including various parts of electronic appliances. At the same time, some problems related to detecting and removing batteries still need solving, such as differences in the design of electronic appliances, occlusions, and the lack of appropriate training data.

In this literature review, datasets for training, research conducted in the field, and current challenges are considered.

2. Existing E-Waste Datasets

2.1 CTSOC E-Waste Dataset

The CTSOC E-Waste Dataset is a publicly available object detection dataset designed for e-waste component recognition. The dataset was developed using images collected from Kaggle repositories and publicly available web sources.

Key Features

- Approximately 990 original images
- Expanded to around 2300 images using data augmentation techniques
- Annotated for object detection tasks
- Includes multiple e-waste categories:
    - Battery
    - Laptop
    - Phone
    - Monitor

Importance to This Research

This dataset is highly relevant because it includes battery annotations within real electronic waste scenarios. The presence of multiple device categories also helps improve model generalization and robustness in mixed e-waste environments.

Limitations

- Limited number of battery-specific images
- Moderate dataset size for deep learning training

- Variability in annotation quality

Dataset Source

- GitHub Repository:
  CTSOC E-Waste Dataset https://github.com/prabindh/ctsoc-ewaste?utm_source=chatgpt.com

2.2 PCB-Vision Dataset

The PCB-Vision Dataset is a benchmark dataset developed for printed circuit board (PCB) analysis using both RGB and hyperspectral imaging.

Key Features

- Multiscene RGB and hyperspectral image data
- Focused on PCB-level component recognition
- Includes electronic components such as:
  - Integrated circuits (ICs)
  - Capacitors
  - Connectors
  - PCB traces

Importance to This Research

Although the dataset is not specifically designed for battery extraction, it provides valuable information for component-level recognition in electronic assemblies. The inclusion of hyperspectral imaging demonstrates the potential of multimodal sensing for improved material and component identification.

Limitations

- Does not focus on mobile phone batteries
- Complex imaging requirements
- Hyperspectral sensors increase system cost

Research Paper

Computer Vision and Pattern Recognition PCB-Vision: A Multiscene RGB-Hyperspectral Benchmark Dataset of Printed Circuit Boards

Dataset Source

PCB-Vision Dataset https://github.com/hifexplo/PCBVision

2.3 Garbage Dataset (GD)

The Garbage Dataset is a large-scale waste classification dataset used for general waste recognition tasks.

Key Features

- Approximately 13,000 images
- Contains multiple waste categories
- Includes battery-related waste samples relevant to e-waste analysis

Importance to This Research

The inclusion of battery-related waste classes provides useful training data for general battery recognition tasks. The large dataset size improves model robustness and supports transfer learning applications.

Limitations

- Primarily designed for waste classification rather than precise object detection
- Limited annotations for extraction-oriented tasks

Research Source

Garbage Dataset Research Article https://arxiv.org/html/2602.10500

2.4 E-Waste Vision Dataset

The E-Waste Vision Dataset was developed for the EWasteNet framework, which applies transformer-based deep learning techniques for e-waste classification.

Key Features

- Contains 8 classes of electronic waste
- Designed for deep learning-based classification
- Supports transformer-based architectures

Importance to This Research

This dataset demonstrates the effectiveness of transformer models in electronic waste recognition and highlights the growing adoption of advanced AI methods for recycling automation.

Limitations

- Primarily focused on classification instead of object localization
- Limited battery extraction-specific annotations

Research Paper

EWasteNet Research Paper https://arxiv.org/pdf/2311.12823

3. Related Research Studies

3.1 Battery Classification Using Computer Vision and OCR

Recent research has demonstrated that combining computer vision techniques with Optical Character Recognition (OCR) can significantly improve battery identification and classification accuracy.

Key Contributions

- Automatic recognition of battery labels
- Battery type classification using textual information
- Improved precision in waste sorting systems

Relevance to This Project

OCR-assisted computer vision can enhance robotic extraction systems by identifying battery specifications, warning labels, and manufacturer information before extraction.

Research Source

Computer Vision and OCR for Battery Classification
https://www.sciencedirect.com/science/article/pii/S2212827122000191

3.2 Robotic Systems for Automated E-Waste Disassembly

Research on robotic e-waste disassembly focuses on automation techniques for safe and efficient removal of electronic components.

Key Contributions

- Automated unscrewing mechanisms
- Robotic manipulation strategies
- Safe dismantling procedures for electronic devices

Relevance to This Project

The study provides insights into integrating robotic arms with computer vision systems for automated battery extraction.

Research Source

Robotic System for Automated Disassembly of Electronic Waste
https://www.researchgate.net/publication/393842955_Robotic_system_for_automated_disassembly_of_electronic_waste_Unscrewing

4. Challenges and Limitations

4.1 Technical Challenges

4.1.1 Lack of Large High-Quality Datasets

One of the major constraints in researching computer vision approaches in e-waste management involves the shortage of good-quality labelled datasets that have been developed for the specific task of battery extraction from the e-waste material.

4.1.2 Difficulty Detecting Small or Hidden Components

Batteries are often partially occluded by device casings, wires, or internal components. Variations in shape, size, and colour further complicate accurate detection.

4.1.3 Real-Time Processing Limitations

Industrial robotic systems require low-latency inference for real-time operation. High-accuracy deep learning models may require substantial computational resources, limiting deployment on embedded robotic platforms.

4.2 Practical Challenges

4.2.1 Variability in Device Design

Electronic devices differ significantly across manufacturers and generations, creating challenges for model generalization.

4.2.2 Safety Concerns During Extraction

Lithium-ion batteries are highly sensitive to puncture, overheating, and mechanical damage. Incorrect robotic manipulation may result in fire or explosion.

4.2.3 Integration Complexity

Combining computer vision, robotic control systems, sensors, and extraction mechanisms into a fully autonomous solution remains technically challenging.

5. Research Gaps

Several research gaps were identified from the reviewed literature:

5.1 Limited Focus on Battery Extraction

Most existing studies focus on general e-waste classification or detection rather than robotic battery extraction specifically.

5.2 Lack of Industrial-Scale Implementations

Few studies demonstrate large-scale deployment of automated e-waste extraction systems in real industrial environments.

5.3 Limited Use of Multimodal Sensing

Most current systems rely only on RGB imaging. The integration of multimodal sensing techniques such as hyperspectral imaging, thermal imaging, and depth sensing remains limited.

6. Conclusion

A review of current literature shows substantial progress towards the implementation of computer vision and robotics for the purposes of managing electronic waste. Current data sets and research contribute to an excellent basis for the development of automated sorting models and component recognition algorithms. Nevertheless, there are a number of issues to be addressed when it comes to the automation of battery-specific extraction operations, including insufficient data sets, speed requirements, and robotic integration problems.

In order to solve some of the aforementioned issues, a customized data set was created for the use of robot batteries extraction.